

EPHRATA, WA, USA

WMO: 727900

Lat: 47.308N

Lon: 119.515W

Elev: 382

StdP: 96.82

Time Zone: -8.00 (NAP)

Period: 01-19

WBAN: 24141

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(1) 12	-12.8	-10.2	-18.5	0.8	-10.5	-15.8	1.0	-8.6	11.2	0.9	9.3	1.2	2.9	340	0.574 (1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(2) 7	14.6	36.6	18.1	34.8	17.7	33.0	17.0	19.4	33.8	18.5	32.4	17.7	31.0	4.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(3) 14.4	10.7	21.7	13.2	9.9	21.6	12.0	9.2	21.8	56.3	34.0	53.4	32.2	50.8	30.7	23.4 (3)

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.0	8.0	DB	-17.3	39.4	3.6	1.6	-19.9	40.6	-22.0	41.6	-24.0	42.5	-26.6	43.7	(4)
(5)			WB	-17.7	20.5	3.4	1.3	-20.2	21.5	-22.2	22.2	-24.2	23.0	-26.7	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-1.1	1.6	6.2	10.5	16.0	20.0	24.7	23.5	18.2	10.7	3.4	-1.3		
	DBStd	9.83	4.61	3.98	3.68	3.52	4.08	4.04	3.59	3.32	3.87	3.91	4.47	4.12		
	HDD10.0	1333	346	236	124	35	3	0	0	0	1	37	202	350		
	HDD18.3	3156	604	469	376	236	97	27	1	2	49	238	449	608		
	CDD10.0	1726	0	0	6	49	189	301	455	419	246	58	3	0		
	CDD18.3	507	0	0	0	1	24	77	198	163	44	1	0	0		
	CDH23.3	6694	0	0	0	25	339	977	2673	2085	574	20	0	0		
	CDH26.7	3164	0	0	0	3	109	417	1404	1016	212	3	0	0		
(14) Wind	WSAvg	3.5	2.8	3.3	3.6	4.0	3.9	4.0	4.0	3.6	3.4	3.1	3.0	2.8		
(15) Precipitation	PrecAvg	202	23	16	16	15	21	19	8	7	6	18	23	31		
	PrecMax	315	66	39	42	31	47	43	47	31	21	65	49	83		
	PrecMin	111	4	0	0	2	2	2	0	0	0	2	1	1		
	PrecStd	52	17	9	11	8	14	12	10	8	6	13	14	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	13.9	19.3	26.4	32.7	37.3	39.0	37.9	34.6	26.2	16.6	10.4		
		MCWB	6.0	8.1	11.0	13.3	15.8	19.0	19.1	18.2	17.2	15.2	11.1	6.7		
	2%	DB	7.6	11.3	17.1	22.8	29.8	33.9	37.0	36.0	31.5	22.3	13.8	7.6		
		MCWB	4.5	6.4	9.5	12.2	15.6	17.7	18.4	18.0	16.3	13.1	8.9	4.9		
	5%	DB	5.8	9.3	14.9	20.2	27.2	31.2	35.4	34.0	29.0	19.9	11.5	5.6		
		MCWB	3.5	5.2	8.2	10.8	14.2	16.5	17.8	17.4	15.4	11.5	7.5	3.3		
	10%	DB	4.0	7.4	13.0	17.8	24.6	28.7	33.7	32.1	26.7	17.7	9.7	3.9		
		MCWB	2.5	4.0	7.2	9.4	13.1	15.3	17.3	16.7	14.3	10.5	6.3	2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	9.1	11.9	14.6	17.3	20.0	20.6	20.9	18.4	16.0	11.6	7.6		
		MCDB	9.4	13.0	17.2	24.7	29.9	34.8	35.5	34.2	31.4	23.1	15.0	9.4		
	2%	WB	4.9	7.0	10.1	12.8	16.0	18.4	19.6	19.0	17.1	13.9	9.9	5.4		
		MCDB	7.2	10.3	15.6	21.4	26.9	32.4	34.5	32.6	29.3	20.5	13.0	7.2		
	5%	WB	3.7	5.7	8.9	11.4	15.0	17.2	18.7	18.1	16.0	12.5	8.3	3.7		
		MCDB	5.4	8.4	13.9	19.3	25.0	29.9	33.4	31.2	27.2	18.2	10.8	5.2		
	10%	WB	2.7	4.6	7.8	10.1	14.0	16.0	17.8	17.3	15.0	11.2	6.9	2.4		
		MCDB	4.0	7.2	12.1	16.7	22.8	27.3	31.8	30.1	25.2	16.5	9.2	3.7		
(35) Mean Daily Temperature Range	MDBR		5.8	8.5	11.4	12.6	13.2	13.1	14.6	14.7	14.2	11.8	8.4	6.0		
	5% DB	MCDBR	8.2	11.2	13.8	15.7	16.2	16.0	16.4	16.7	17.2	14.5	10.6	8.4		
		MCWBR	5.8	7.2	7.9	7.6	6.3	6.1	6.1	6.3	7.2	7.4	6.8	6.0		
	5% WB	MCDBR	7.4	10.2	13.0	15.2	14.5	15.2	15.4	15.5	15.7	12.9	9.7	7.4		
		MCWBR	5.5	6.9	7.9	7.8	6.1	6.1	6.1	6.1	6.9	7.2	6.6	5.5		
(40) Clear-Sky Solar Irradiance	taub		0.277	0.283	0.294	0.318	0.333	0.335	0.336	0.334	0.322	0.303	0.288	0.273		
	taud		2.506	2.559	2.572	2.473	2.446	2.467	2.463	2.460	2.496	2.544	2.545	2.515		
	Ebn at Noon		829	903	941	938	930	927	921	910	893	862	808	790		
	Edh at Noon		62	73	83	102	108	107	106	102	89	73	60	55		
(44) All-Sky Solar Radiation	RadAvg		1.08	2.21	3.52	5.21	6.36	6.97	7.57	6.33	4.67	2.87	1.44	0.90		
	RadStd		0.13	0.28	0.24	0.32	0.39	0.39	0.30	0.35	0.26	0.24	0.16	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	-0.44	N/A	N/A	N/A	+95	+47

Nomenclature: See separate page